

Buy now, pay later, underwritten by verified identity instead of collateral.

Live kudira-fi.vercel.app Repo github.com/shrooms08/kudira Demo youtu.be/XS926c6qZ8g

PROBLEM

Buy now, pay later works because the lender knows who you are. Miss a payment and it follows you. On an anonymous chain nothing follows you, so a defaulter opens a new wallet and starts again.

That is why on-chain lending converged on overcollateralization: deposit 150 to borrow 100. It is not a design preference, it is what is left when you cannot identify a borrower. A loan secured by more than the loan is worth is a pawnshop, not credit. If you already had the 150, you would not need the loan. BNPL is the one consumer credit primitive that cannot exist without identity: the buyer receives goods before paying, and the only thing securing the loan is an identity they cannot discard.

SOLUTION

A buyer checks out. Kudira reads their Cleanverse credential and gets back a tier, a standing and an expiry, never a name, a document or an account number. That standing is the credit limit. The merchant is paid in full **in the same transaction as origination**, so they carry no credit risk and wait for nothing. The buyer approves once and each installment is auto-debited on its due date.

The grade is the A-Pass `subTier`, an integer 0 to 99 held on-chain. **Limit = subTier × 10**. An on-time installment is +5, a default is -20, both saturating. Below 10 no new plans originate. No plan may extend past the credential's expiry.

A default lowers Kudira's own grade. It does not freeze the credential. Freezing would lock the borrower out of every service on the Cleanverse network including the ability to repay us. A lender that blocks repayment has mispriced its own incentives. Recovery is reputational rather than collateral-based: the record follows the person, not the wallet, and is visible to every protocol reading the same A-Pass.

CVI · CVA INTEGRATION POINTS

Every claim below is a public read on Base Sepolia.

CVI, the underwriting input

`query_apass` returns tier, subTier, standing and expiry. The credential determines the limit and the schedule. Repayment behaviour is written back to subTier via `generate_apass` with `override`, so the grade travels with the person rather than living privately in our database. Proven on-chain: 50 → 55 → 60 → 65 → 70 across four collected installments.

CVA, the settlement asset

Settlement is in KUSDC, a CVA issued under our Issue Member scope through the documented Launch CVA path, after Cleanverse's Base faucet stopped dispensing aUSDC on 24 July. It carries the same `min_tier 5` rule as aUSDC and is gated identically on **both** sides of every transfer, inside `_update`, before any balance check. We can mint KUSDC. We cannot mint past a credential check. The gate is Cleanverse's, not ours.

CCP, the compliance gate

KudiraPool is registered with the CVI Compliance Validator as a Pattern 2 single-contract lending pool. `complianceVerify(pool, wallet)` is read **on-chain** at origination alongside the REST `validator/verify` call, and the dashboard shows both answers for three wallets, one generated at the moment the button is pressed. If the two ever disagree the UI says so rather than picking one. `getRulesV2(pool)` returns `(0x0000, 0x0000, 5, 0, 0)` and `isRegistered(pool)` is true, so our policy is stored in Cleanverse's contract, not asserted by us.

Where the integration ends, and why

Travel Rule reporting is bound to Cleanverse's indexed settlement flow. We proved a real customer-to-customer aUSDC transfer generates a report, and that a mint from `address(0)` never will because it has no originator. KUSDC sits outside the index, so the dashboard shows the live `TR_001` response and explains it rather than hiding it.

DEPLOYED CHAINS

KudiraPool	0x4a898781AFAd85BE7103126952BcBbFCCC24199e	MerchantRegistry	0x05e2A2473e710435484f6B3b288677618E95bB15
CreditLine	0x49cd6c00EC00116Eed598b8e07f1B0D7A4805cBE	KUSDC (CVA)	0x036BCFeB3cfE93dfc47f5A935D7f663b99ACab1E
InstallmentPlan	0xb4c055e7e880A684F92764358Dc12d25577d39D8	CCP validator	0xaC7e5179C2C7f03f209136886c172eb34F161792

Base Sepolia, chainId 84532. 116 Foundry tests. Separation of duties on-chain: the operator can create debt but cannot add a payee or withdraw liquidity, and owner and operator are different keys. Contracts were also verified on Monad testnet before we moved to Base.

SCALABILITY

Kudira is merchant-agnostic by construction. `MerchantRegistry` supports arbitrary merchants and origination takes the merchant as a parameter, so adding a shop is a credential plus a registry write, not a redeploy. Installment count and cadence are origination parameters rather than hardcoded product decisions, and a down payment at checkout is the same call with the first installment settled immediately, which is how first-time exposure gets bounded in production.

What scales next: an indexer in place of `planCount` iteration, self-serve merchant onboarding, per-merchant dashboard authentication, and a liquidity layer where third-party lenders fund the pool and earn on repayments. A first-time borrower's limit is currently seeded from their credential at origination, so the underwriting preview should read the A-Pass directly rather than the on-chain account.

Take CVI out and the limit has to be inferred from wallet history anyone can manufacture, one person opens unlimited wallets, and default costs a discarded keypair. Removing Cleanverse does not degrade Kudira. It deletes it.